

AI-Driven Virtual Patient Simulations for Communication Training in Medical Students: A Pilot Study

Beatriz Zambrano-Serrano

MetaMedics VR

Raquel Cueto-Galán

rcueto@uma.es

University of Malaga

Alejandro Escamilla-Sánchez

Faculty of Medicine, University of Malaga

Maria del Rocio Ramirez-Rodríguez

University of Malaga

Nuria Sanchez-Corrales

University of Malaga

Andres Fontalba-Navas

University of Malaga

Research Article

Keywords: Virtual Patients, Natural Language Processing (NLP), Clinical Training, Objective Structured Clinical Examination (OSCE), AI-Driven Feedback, Human-AI Interaction

Posted Date: October 27th, 2025

DOI: <https://doi.org/10.21203/rs.3.rs-7270436/v1>

License:  This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: No competing interests reported.

Abstract

Background

Effective communication is essential in medical practice, yet traditional training methods face limitations, including resource constraints, variability in instructional quality, and inconsistent feedback. Artificial Intelligence (AI)-driven tools provide scalable and interactive solutions for developing communication skills in a structured manner. This pilot study evaluated the feasibility and effectiveness of an AI-based virtual patient simulation tool for enhancing communication skills in medical students. Specifically, the study examined whether the tool would demonstrate high usability among participants and whether the AI-generated feedback would be perceived as clear, well-structured, and useful for the development of communication competencies.

Methods

A total of 102 medical students from the University of Málaga participated in a GPT-4-powered virtual patients' interactions across two standardized clinical cases. Usability, satisfaction, and feedback quality were assessed via post-session surveys, Likert-scale ratings, and qualitative feedback. Statistical analyses included a binary logistic regression analysis that was conducted to examine whether gender influenced the perceived need for prior training before using the tool.

Results

Binary logistic regression revealed that gender was a significant predictor of perceived need for prior training. Women were significantly less likely than men to consider such training necessary ($\beta = -1.24$, $OR = 0.29$, $SE = 0.47$, $z = -2.63$, $p = 0.008$), indicating a 71% reduction in odds. Most participants (60.8%) reported no need for prior training, and 86% found the tool easy or too easy indicating high intuitive usability.

Conclusions

Findings confirm the feasibility, high usability, and effectiveness of AI-driven virtual patient simulations, with significant differences in perceived feedback quality across dimensions. Further research should explore longitudinal skill retention, hybrid models integrating human-led debriefing, and AI-enhanced nonverbal communication analysis to refine AI-driven communication training, while addressing ethical concerns such as bias, misinformation, and the limitations of AI in providing emotionally contextualized feedback.

Introduction

Effective communication is fundamental to medical practice, playing a crucial role in building patient trust, ensuring accurate diagnosis, and promoting adherence to treatment plans. It serves as the cornerstone of patient-centered care, not only are clinical decision-making influenced but also patient

satisfaction and overall health outcomes [1]. Strong communication skills enhance the patient experience as well as contribute to safer and more efficient healthcare delivery [2, 3] By fostering an open and supportive dialogue, effective communication enables patients to express their concerns more freely, improving the accuracy of the diagnostic process and boosting shared decision-making. Moreover, active listening and conveying comprehensible information strengthens the doctor-patient relationship and reduces medical errors [4]

Consequently, developing communication skills is a core component of medical education. Traditional methods for teaching these skills, such as role-playing with peers or standardized patients, supported by lectures and feedback, are effective but face some challenges [5, 6]. These approaches are resource-intensive, time-consuming, and prone to variability in quality, limiting their scalability and consistency [7] Additionally, the effectiveness of these methods often depends on the availability of trained facilitators and standardized patients, which may not be feasible in all educational settings. As the demand for scalable and standardized training grows, there has been a shift from traditional standardized patients to AI-driven virtual patient simulations, which offer consistent, cost-effective, and structured learning experiences [8, 9].

Recent advances in AI have introduced scalable solutions to address limitations in traditional medical communication training, particularly in ensuring consistency and real-time feedback [10]. AI-driven virtual patient simulations are able to create interactive learning environments that mimic diverse clinical scenarios with a high degree of realism, enabling learners to repeatedly practice complex conversations in a low-risk setting [11]. Unlike traditional role-play methods, AI-based simulations ensure consistency in patient interactions and feedback, thereby reducing the variability seen in human-led training [12]. However, research remains limited regarding AI's applicability in diverse linguistic and educational settings, especially within Spanish-speaking medical education contexts [13]. AI-based teaching role plays have emerged as a promising alternative, offering enhanced learning experiences for students in health study programs [14]

Integrating AI into pre-clinical education has been identified as a crucial step toward enhancing medical training, as AI-driven tools provide interactive, adaptive, and personalized learning environments that traditional teaching methods may lack [15]. AI-driven platforms can dynamically adjust difficulty levels, simulate emotionally charged interactions, and track longitudinal progress, offering a more tailored and evidence-based approach to communication skills training [16]. Some studies have even highlighted the benefits of AI chatbots in cognitive training, demonstrating their effectiveness in improving executive function and enhancing educational engagement [17].

Furthermore, advancements in AI-driven conversational agents have shown promise in healthcare settings, including their role in clinical communication training [18]. Integrating text-based chatbot-driven simulations or more advanced voice-recognition-based AI simulations into medical education might further expand opportunities for students to refine their communication skills in a structured yet flexible manner, bridging the gap between theoretical knowledge and real-world patient interaction [13]

This is evident in current literature. Previous studies emphasized AI's ability to simulate realistic patient interactions, analyze communication performance, and provide immediate feedback [12]. Similarly, other authors demonstrated that GPT-4-based virtual patients offer medically plausible, structured responses with feedback accuracy comparable to human evaluators [13]. Collectively, these studies underscore AI's promise in developing communication skills while reducing the resource demands of traditional methods.

In spite of these developments, important areas of concern remain unaddressed. Many studies are descriptive and limited in generalizability due to small sample sizes used or narrowly defined clinical scenarios. Holderried identified discrepancies in the specificity of AI feedback, particularly for nuanced categories [13]. Nagi observed that most AI applications in medical education target technical and diagnostic tasks, leaving a gap in their use for soft skills (e.g., communication) [15]. Additionally, challenges persist in replicating human interaction complexity, including nonverbal communication and empathy—critical components of effective patient care.

Unlike previous research emphasizing technical performance or single-use scenarios, this study evaluates key factors such as student satisfaction, engagement, and the tool's educational value. By incorporating two standardized cases, the research ensures consistency while simulating diverse clinical interactions, providing practical insights into the tool's applicability in real-world educational settings. Additionally, the study explores the potential of AI-based simulations to complement traditional teaching methods, forming a hybrid approach that combines the scalability and consistency of AI with the depth and nuance of in-person learning [19–22]

The purpose of this study is to assess the usability, effectiveness, and acceptability of an AI-based virtual patient simulation tool for communication training among medical students at the University of Málaga. By evaluating its impact on structured history-taking, user satisfaction, and educational value, the study aims to provide insights into the integration of AI technologies in medical education. Additionally, it seeks to identify the strengths and limitations of AI-driven tools in addressing key challenges such as scalability, variability, and student engagement, ultimately contributing to the development of innovative and effective training methodologies.

Methods

This was a cross-sectional, exploratory pilot study with post-intervention evaluation using mixed methods (quantitative surveys and qualitative open-ended responses). No control group was employed. This pilot study evaluated the usability, effectiveness, and acceptability of an AI-based virtual patient simulation tool for communication skills training among medical students at the University of Málaga. The study followed a controlled, in-person training design using standardized cases to ensure consistency and minimize variability in participant experience. The primary objective was to assess the tool's impact on structured history-taking, user satisfaction, and educational value while identifying strengths and limitations related to scalability, variability, and student engagement. A self-developed

questionnaire was designed specifically for this study to collect data on. The full English version of the questionnaire is provided as Supplementary File 1.

The study sample comprised 102 medical students from the University of Málaga voluntarily participated and informed consent was obtained before engagement. Eligibility criteria included active enrollment in the medical program and demonstrated fluency in Spanish, as the training sessions were delivered exclusively in this language. No exclusions were made based on gender, academic performance, or prior experience with AI technologies. The study's independent variable was interaction with the AI-based virtual patient tool, while dependent variables included different aspects of how participants evaluate the AI-based tool, being: (1) Usability: ease of use, technical issues, training requirements; (2) Perceived realism: naturalness of interactions, clinical relevance; (3) Feedback quality: clarity, usefulness, comparison to human feedback; (4) Overall satisfaction: confidence in learning, preference for hybrid models.

The AI tool used Natural Language Processing (NLP) to simulate patient interactions, allowing students to conduct dynamic, structured medical consultations. The system was powered by GPT-4, a large-scale transformer-based language model fine-tuned for medical dialogue simulation. The AI-generated responses were trained on a mix of publicly available medical communication datasets, standardized patient dialogues, and domain-specific content curated by medical educators. Additionally, prompt engineering techniques were employed to ensure structured patient responses and guided feedback aligned with established clinical training frameworks. The AI provided real-time, text-based responses and delivered structured feedback on communication effectiveness.

To enhance clinical relevance, the virtual patient model incorporated predefined rule-based structures for differential diagnosis and probabilistic scoring of student responses, guaranteeing that feedback aligned with expected clinical reasoning. However, the AI system had notable limitations, including the inability to interpret nonverbal cues, emotional tone, or nuanced contextual information. Future iterations may integrate multimodal AI capabilities, such as speech recognition and sentiment analysis, to improve response accuracy and enhance emotional engagement.

Each student completed two AI-driven interactions based on standardized clinical cases following a structured consultation framework, which included: (1) greeting and patient introduction, (2) chief concern and history of present illness, (3) general health indicators review, (4) past medical history, and (5) family and medication history.

Data collection was conducted immediately after each training session via an anonymous structured questionnaire. Participants provided Likert-scale ratings on usability, realism, and feedback quality, while open-ended questions allowed for qualitative feedback on strengths, weaknesses, and areas for improvement. To minimize recall bias, surveys were completed immediately post-session, with trained observers ensuring response consistency.

Students were informed that participation was voluntary, responses were anonymous, and they could withdraw at any time without academic consequences. Data were de-identified before analysis to ensure confidentiality.

Data were analyzed using SPSS software, with statistical significance set at $p < 0.05$. Chi-Square Tests were used to assess the distribution of categorical usability ratings (e.g., “easy,” “very easy,” “neutral or difficult”). One-Sample T-Tests compared mean usability and feedback ratings against a neutral threshold of 5 on a 0–10 scale. One-Way ANOVA was performed to determine whether statistically significant differences existed across feedback dimensions, including clarity, usefulness, and consultation strategies.

Results

The study included 102 medical students from the University of Málaga, of whom 73.5% were female ($n = 75$), and 85.3% ($n = 87$) were in their fourth year of study. All participants were fluent in Spanish, as required for training. Regarding prior experience, 78.4% ($n = 80$) had no clinical interview experience, and 95.1% ($n = 97$) had never used a similar AI-based communication training tool before (Table 1).

Table 1
Demographic Information

Characteristic	Value
Total Participants	102
Gender (Female) (%)	73.5%
Gender (Male) (%)	25.5%
Year of Study	1st to 6th Year
Year of Study (Most Common, 4th)(%)	85.3%
Year of Study (Second Most Common, 5th) (%)	9.8%
Age (Most Common, 21) (%)	52%
Age (Second Most Common, 22) (%)	24%
Inclusion of International Students	Yes

In terms of usability, 86% ($n = 88$) of participants rated the AI-based tool as “easy” or “very easy” to use, while 14% ($n = 14$) found it neutral or difficult to use. Participants evaluated AI-generated feedback using a 0–10 scale. As shown in Table 2, the clarity of feedback was rated 8.2 (SD = 2.1, median = 9), and overall satisfaction with AI-generated feedback was 7.4 (SD = 2.5, median = 8). The highest-rated dimensions were history-taking guidance (mean = 8.3, SD = 1.8, median = 9) and consultation strategies (mean = 8.15, SD = 2.04, median = 9). When comparing AI feedback to human tutor feedback, participants rated AI feedback’s effectiveness as comparable to traditional feedback (mean = 8.38, SD =

2.10, median = 9), while AI's ability to fully replace human feedback was rated lower (mean = 6.14, SD = 3.03, median = 6), indicating a higher variability in participant opinions (SD = 3.04).

Table 2
Summary Statistics for Each Statement

Statement	Mean	Median	Sd
"Overall, I am satisfied with the feedback received."	7.4	8	2.5
"The information provided was clear, and I understood it without problems."	8.2	9	2.3
"The feedback offers an accurate and precise evaluation of my interaction with the simulated patient."	7.5	8	2.6
"The feedback correctly addresses the essential aspects of the clinical interview and the specific diagnostic metrics."	8.1	9	2.2
"The feedback was useful for improving how I present myself and initiate communication with the patient."	8.2	9	2.0
"The feedback was useful for identifying the reason for consultation and exploring the patient's current condition in depth."	8.3	9	1.8
"The feedback has provided me with useful strategies to explore the patient's family and personal history."	8.3	9	1.9
"I believe the feedback helps improve my approach to consulting with a patient."	8.3	9	1.9
"The feedback has been useful in identifying areas of the clinical interview I need to improve."	7.64	8	2.47
"I feel more prepared to conduct clinical interviews with real patients."	7.82	8	2.21
"The feedback provided by the AI is more effective in acquiring communication skills."	8.15	9	2.04
"The feedback provided by the AI is more efficient, allowing for the acquisition of these skills more easily, with fewer costs and resources."	6.79	8	2.64
"The feedback provided by the AI does not replace the experience offered by clinical tutors; I prefer to receive their feedback in the future."	6.14	7	3.03
"The feedback provided by the AI does not replace the experience offered by clinical tutors; I prefer to receive their feedback in the future."	6.27	7	3.04
"The feedback provided by the AI offers the same information (in terms of quality and depth) as a clinical tutor."	8.38	9	2.10
"Both training through AI and real clinical practice are useful, so both should be offered complementarity."	8.34	9	2.52

Qualitative responses (n = 33) were analyzed for recurring themes. Participants most frequently mentioned "realism," "interaction," and "feedback depth" as positive aspects. The most common areas

for improvement were “time” (11 responses), “adaptability” (7), and “case diversity” (6). Regarding least useful aspects, 29 participants responded with “no” or “none”, indicating that a majority did not identify a major drawback. When asked about AI’s ability to assess nonverbal communication, 10 participants mentioned body language, 6 referenced nonverbal cues, and 7 mentioned empathies as an area for further refinement.

A binary logistic regression analysis was performed to assess whether gender influenced students perceived need for preparatory training prior to using the AI-based application. The analysis revealed that female participants were significantly less likely than their men counterparts to consider such training necessary. Specifically, being a woman was associated with a 71% decrease in the odds of reporting a perceived need for prior instruction ($\beta = -1.24$, OR = 0.29, SE = 0.47, $z = -2.63$, $p = 0.008$). This finding suggests that compared to men, women were more likely to perceive the tool as intuitive or accessible, requiring minimal orientation before use (Table 3).

Table 3
Results of the binary logistic regression model predicting gender based on the perception of the need for prior training in using the application.

Parameter	Estimate (log-odds)	Standard Error	Z value	p-value	OR (odds ratio)	p-value (signif.)	Interpretation
(Intercept)	1.6487	0.3453	4.775	0.0000018	5.2	0.000002 ***	Intercept: Baseline probability of being a WOMAN when training is not needed
Needs_ Training = Yes	-1.2432	0.4726	-2.63	0.00853	0.288	0.00853 **	When training is considered necessary, the odds of being a WOMAN decrease

In open-ended qualitative responses, participants emphasized realism, structured feedback, and patient interaction quality. When asked about AI’s ability to evaluate nonverbal communication, 10 participants specifically mentioned body language as an area needing improvement, while 6 participants highlighted the need for enhanced nonverbal cue assessment, and 7 cited empathy as a component requiring refinement.

Discussion

The findings of this pilot study provide preliminary support for the integration of AI-driven virtual patient simulations as a complementary tool in the development of communication competencies among medical students. Participants rated the tool highly for usability, clarity, and structured feedback,

reinforcing AI's potential as a scalable and interactive training method. These results are consistent with those reported in previous studies, showcasing that AI-driven virtual patient simulations can enhance communication skills through standardized, repeatable clinical encounters that provided structured feedback [12]. Similarly, Kyaw et al. (2019) reported that digital training tools can be as effective as traditional methods when well-integrated into curricula [10]. The low requirements for technical support and prior training suggests that AI-based simulations possess favourable scalability in medical education without extensive onboarding resources. This aligns with Chan et al. (2019) who emphasized that AI adoption is easier when tools require minimal onboarding [23]. Additionally, others authors highlights AI's potential in pre-clinical training, allowing students to develop communication skills before clinical exposure [24]. Findings also suggest that AI-generated feedback is particularly valuable in history-taking and consultation guidance, but human-led instruction remains essential for adaptive, emotionally nuanced, and culturally competent training.

However, the analysis identified significant differences across feedback categories, indicating that some aspects of feedback were perceived as more useful than others. This is important as not to waste students' time and resources on aspects that do not provide value. For example, structured real-time AI feedback can improve learning outcomes whereas other types of feedback provided by AI may not be as insightful [25]. Challenges remain in feedback personalization. While AI-generated feedback was rated comparable to human tutor feedback its ability to fully replace human feedback was significantly lower. This suggests that AI alone is insufficient and should complement human-led instruction. Hamilton (2024) highlighted that AI lacks the empathic responsiveness and contextual awareness needed for high-stakes clinical communication training, reinforcing the importance of human oversight [26, 27].

AI-driven simulations present scalable training advantages, but ethical concerns must be addressed for responsible implementation. A primary concern is bias in AI-generated feedback, as AI models are trained on pre-existing datasets that may reflect disparities in medical education and communication. If unchecked, biases could reinforce disparities in clinical training and patient interactions [28] Another critical issue is AI-generated misinformation, where models may provide incorrect responses if not carefully curated [26]. This remarks the need for continuous human oversight. Additionally, the lack of transparency in AI decision-making raises concerns about interpretability—students may accept AI feedback without questioning its validity, which could impact clinical reasoning development [29]. Future studies should explore ethical AI design principles to enhance fairness, accuracy, and accountability in AI-driven training

A key contribution of this study is its focus on a Spanish-speaking cohort, addressing a gap identified by Werner et al. (2024), who noted that most AI-driven medical education research is conducted in English-speaking populations [31]. Medical communication styles, cultural norms, and language structures vary significantly, potentially influencing how students interact with AI-driven virtual patients.

While AI-based training demonstrates strong applicability, its generalizability across diverse linguistic and cultural settings remains uncertain. AI-generated feedback may not align with culturally specific

communication norms, which could impact the effectiveness of training across different educational contexts.

Several limitations must be considered. Participants identified the limited variety of clinical scenarios and the system's inability to assess nonverbal communication cues as notable limitations that constrain the ecological validity of the simulation experience. These findings corroborate the restricted ability of AI to assess nonverbal behavior and emotional dynamics as a key drawback in simulated patient interactions [32]. AI-driven training tools currently rely on text-based interactions, reducing their ability to assess tone, facial expressions, and nonverbal cues. Additionally, given that most participants had never used a similar AI-based training tool, the high usability ratings suggest that AI-driven simulations may be well-suited for early-stage communication training. This supports that AI-assisted training bridges the gap between theoretical knowledge and practical application [33]. However, for more advanced learners, AI-driven feedback alone may be insufficient, necessitating hybrid models that integrate AI guidance with human mentoring [34]. Another major limitation is the potential for biases in AI-generated responses. AI models are trained on structured datasets, which may not reflect the full diversity of clinical cases or patient interactions. Ethical concerns around bias, misinformation, and AI transparency must be addressed before AI-driven simulations are widely implemented in medical education [27, 29].

Future research should evaluate long-term skill retention to determine whether AI-driven training translates into sustained improvements in clinical practice. Comparative studies between AI-generated and human-led feedback could provide deeper insights into differences in realism, effectiveness, and knowledge retention, refining hybrid training models [34]. Additionally, addressing AI's limitations in interpreting nonverbal cues is crucial [32]. The integration of speech recognition and sentiment analysis could enhance realism and improve feedback accuracy in high-stakes medical communication [29]. The role of prompt engineering in optimizing AI-driven training also deserves attention, as structured prompts and feedback loops significantly impact educational outcomes [35]. Beyond communication training, AI applications in surgical education and mental health training offer promising opportunities for procedural learning and clinical decision-making [36, 37]. To enhance global applicability, future research should explore AI-driven communication training across diverse linguistic and cultural settings. Comparative studies in multilingual medical cohorts could assess how AI-generated feedback aligns with cultural variations in medical communication, informing best practices for AI adaptation in global medical education [31].

By addressing these pedagogical, ethical, and technological challenges, AI-driven simulations can evolve into a more effective, context-aware, and equitable training tool, ultimately enhancing medical education and patient care outcomes.

Conclusion

This study underscores the considerable potential of AI-driven virtual patient simulations as innovative tools for enhancing communication skills training in medical education. Their ability to deliver scalable,

structured, and intuitively accessible feedback positions them as a valuable complement to traditional pedagogical methods particularly for learners in the early stages of clinical development. Nevertheless, significant limitations remain, notably the current inability of AI systems to interpret nonverbal cues, cultural adaptability, and deliver fully personalized feedback. Moreover, ethical concerns related to algorithmic bias, potential misinformation, and lack of transparency further highlight the need for human oversight and responsible AI design. Future research should focus on hybrid training models, long-term skill retention, and cross-cultural applicability to ensure AI tools are both effective and equitable across diverse educational contexts.

Abbreviations

AI

Artificial Intelligence

OSCE

Objective Structured Clinical Examination

NPL

Natural Language Processing

Declarations

Ethics statement

This study was conducted in accordance with the principles of the Declaration of Helsinki. As it involved standardized virtual patients for the development and technical validation of an interactive training tool, ethical approval was not required, as confirmed by the Provincial Research Ethics Committee of Málaga (CEIM), in accordance with local legislation and institutional requirements. Written informed consent was obtained from all participants. Social media data were accessed and analyzed in compliance with the platform's terms of use and all applicable regulations.

Consent for publication

Not applicable

Availability of data and materials

The data that support the findings of this study are not openly available due to reasons of sensitivity and confidentiality. However, they are available from the corresponding author upon reasonable request.

Competing interests

The primary author declares that she is affiliated with MetaMedicsVR, which developed the AI-based virtual patient tool used in this study. The University of Málaga led the study as an academic partner in the project. There are no additional conflicts of interest to disclose.

Funding:

This study received no external funding. MetaMedicsVR provided access to the simulation tool, but the research was conducted independently by the University of Málaga without financial sponsorship

Authors' contributions

BZS: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. RC-G: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. AE-S: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. MSC: Conceptualization, Data curation, Formal analysis, Writing – review & editing. NSC: Conceptualization, Data curation, Formal analysis, Writing – review & editing. AF-N: Investigation, Validation, Writing – review & editing, Conceptualization, Funding acquisition, Supervision, Methodology, Writing – original draft, Formal analysis, Software, Visualization, Resources, Data curation, Project administration.

Acknowledgments:

We would like to extend our gratitude to the University of Málaga for their collaboration and support throughout this project. Special thanks are due to the volunteers.

References

1. Ortiz-Paredes D, Henet PA, Desseilles M, Rodríguez C. Empathy in family medicine postgraduate education: A mixed studies systematic review. *Med Teach*. 2025;47(2):275–91. <https://doi.org/10.1080/0142159X.2024.2328324>.
2. Stewart MA. Effective physician-patient communication and health outcomes: a review. *CMAJ*. 1995;152(9):1423–33. <https://pubmed.ncbi.nlm.nih.gov/7728691/>.
3. Leonard M, Graham S, Bonacum D. The human factor: the critical importance of effective teamwork and communication in providing safe care. *Qual Saf Health Care*. 2004;13(Suppl 1):i85–90. <https://pubmed.ncbi.nlm.nih.gov/15465961/>.
4. Mackintosh NJ, Davis RE, Easter A, Rayment-Jones H, Sevdalis N, Wilson S, et al. Interventions to increase patient and family involvement in escalation of care for acute life-threatening illness in community health and hospital settings. *Cochrane Database Syst Rev*. 2020;12:CD012829. <https://doi.org/10.1002/14651858.CD012829.pub2>.
5. Bosse HM, Nickel M, Huwendiek S, Jünger J, Schultz JH, Nikendei C. Peer role-play and standardised patients in communication training: a comparative study on the student perspective on

- acceptability, realism, and perceived effect. *BMC Med Educ.* 2010;10:27.
<https://doi.org/10.1186/1472-6920-10-27>.
6. Xu L, Liu W, Jiang X, Li Y. Impact of using peer role-playing on the clinical skills performance of pediatric trainees. *BMC Med Educ.* 2023;23:654. <https://doi.org/10.1186/s12909-023-04554-0>.
 7. Novais F, Ganança L, Barbosa M, Telles-Correia D. Communication skills in psychiatry for undergraduate students: A scoping review. *Front Psychiatry.* 2022;13:972703.
<https://doi.org/10.3389/fpsyt.2022.972703>.
 8. Cook DA, Overgaard J, Pankratz VS, Del Fiore G, Aakre CA. Virtual Patients Using Large Language Models: Scalable, Contextualized Simulation of Clinician-Patient Dialogue With Feedback. *J Med Internet Res.* 2025;27:e68486. <https://doi.org/10.2196/68486>.
 9. Komazawa N, Yokohira M. Simulation-based education in the artificial intelligence era. *Cureus.* 2023;15(6):e40940. <https://doi.org/10.7759/cureus.40940>.
 10. Kyaw BM, Posadzki P, Dunleavy G, Semwal M, Divakar U, Matchar DB, et al. Digital education for health professions on communication skills: Systematic review and meta-analysis by the Digital Health Education Collaboration. *J Med Internet Res.* 2019;21(8):e12967.
<https://doi.org/10.2196/12967>.
 11. Gazquez-Garcia J, Sánchez-Bocanegra CL, Sevillano JL. AI in the health sector: Systematic review of key skills for future health professionals. *JMIR Med Educ.* 2025;11:e58161.
<https://doi.org/10.2196/58161>.
 12. Stamer T, Steinhäuser J, Flägel K. Artificial intelligence supporting the training of communication skills in the education of health care professions: Scoping review. *J Med Internet Res.* 2023;25:e43311. <https://doi.org/10.2196/43311>.
 13. Holderried F, Stegemann-Philipps C, Herrmann-Werner A, Festl-Wietek T, Holderried M, Eickhoff C, et al. A language model-powered simulated patient with automated feedback for history taking: Prospective study. *JMIR Med Educ.* 2024;10:e59213. <https://doi.org/10.2196/59213>.
 14. Schliz K, Meindl B, Rädcl-Abläss K, Schlick C. Teaching opportunities for anamnesis interviews through AI-based teaching role plays: A survey with online learning students from health study programs. *BMC Med Educ.* 2025. <https://doi.org/10.1186/s12909-025-06756-0>.
 15. Nagi F, Salih R, AlZubaidi M, Shah H, Alam T, Shah Z, et al. Applications of artificial intelligence (AI) in medical education: A scoping review. *Healthcare Transformation with Informatics and Artificial Intelligence.* IOS; 2023. <https://doi.org/10.3233/SHTI230581>.
 16. Pergantis P, Bamicha V, Skianis C, Drigas A. AI chatbots and cognitive control: Enhancing executive functions through chatbot interactions: A systematic review. *Brain Sci.* 2025;15(1):47.
<https://doi.org/10.3390/brainsci15010047>.
 17. Chew HSJ. The use of artificial intelligence-based conversational agents (chatbots) for weight loss: Scoping review and practical recommendations. *JMIR Med Inf.* 2022;10(4):e32578.
<https://doi.org/10.2196/32578>.

18. Xue J, Zhang B, Zhao Y, Zhang Q, Zheng C, Jiang J, et al. Evaluation of the current state of chatbots for digital health: Scoping review. *J Med Internet Res*. 2023;25:e47217. <https://doi.org/10.2196/47217>.
19. Jackson P, Ponath Sukumaran G, Babu C, Tony MC, Jack DS, Reshma VR, et al. Artificial intelligence in medical education - perception among medical students. *BMC Med Educ*. 2024;24(1):804. <https://doi.org/10.1186/s12909-024-05760-0>.
20. Civaner MM, Uncu Y, Bulut F, Chalil EG, Tatli A. Artificial intelligence in medical education: a cross-sectional needs assessment. *BMC Med Educ*. 2022;22(1):772. <https://doi.org/10.1186/s12909-022-03852-3>.
21. Salih SM. Perceptions of Faculty and Students About Use of Artificial Intelligence in Medical Education: A Qualitative Study. *Cureus*. 2024;16(4):e57605. <https://doi.org/10.7759/cureus.57605>.
22. Rani S, Kumari A, Ekka SC, Chakraborty R. Perception of Medical Students and Faculty Regarding the Use of Artificial Intelligence (AI) in Medical Education: A Cross-Sectional Study. *Cureus*. 2025;17(1):e77514. <https://doi.org/10.7759/cureus.77514>.
23. Chan KS, Zary N. Applications and challenges of implementing artificial intelligence in medical education: Integrative review. *JMIR Med Educ*. 2019;5(1):e13930. <https://doi.org/10.2196/13930>.
24. Pohn B, Mehnen L, Fitzek S, Choi KEA, Braun RJ. Integrating artificial intelligence into pre-clinical medical education: Challenges, opportunities, and recommendations. *Front Educ*. 2025;10:1570389. <https://doi.org/10.3389/feduc.2025.1570389>.
25. Lee A, Hwang J, Kim S, Park H. Artificial intelligence-powered virtual patient simulations for medical training: A systematic review. *Med Educ*. 2020;54(12):1038–52. <https://doi.org/10.1111/medu.14152>.
26. Hamilton A. AI-driven feedback in clinical communication training: Strengths, limitations, and future directions. *Cureus*. 2024;16(5):e233783. <https://doi.org/10.7759/cureus.233783>.
27. Hamilton A, Molzahn A, McLemore K. The evolution from standardized to virtual patients in medical education. *Cureus*. 2024;16(5):e300313. <https://doi.org/10.7759/cureus.300313>.
28. Lu Y, Zhao X. Evaluating AI-based virtual patients for clinical communication training: A systematic review. *Med Teach*. 2022;44(7):654–72. <https://doi.org/10.1080/0142159X.2022.2036423>.
29. Indolfi C, Klain A, Dinardo G, Decimo F. Artificial intelligence in the transition of allergy: A valuable tool from childhood to adulthood. *Front Med*. 2024;11:1469161. <https://doi.org/10.3389/fmed.2024.1469161>.
30. Werner M, Petri H, Schmitt B. The role of AI-driven mobile health applications in medical education: A systematic review. *JMIR Mhealth Uhealth*. 2021;9(12):e31559. <https://doi.org/10.2196/31559>.
31. Werner M, Lang B, Köhler C. The role of AI in non-English medical education: A systematic review of existing literature. *BMC Med Educ*. 2024;24:146. <https://doi.org/10.1186/s12909-024-04172-5>.
32. Lu H, Chen J, Liang F, Tan M, Zeng R, Hu X. Understanding Emotional Body Expressions via Large Language Models. *arXiv [Preprint]*. 2024. arXiv:2412.12581. <https://arxiv.org/abs/2412.12581>

33. Shorey S, Ang E, Yap J, Ng ED, Lau ST, Chui CK. Virtual reality simulation in nursing education: A systematic review. *Nurse Educ Today*. 2019;79:134–42. <https://doi.org/10.1016/j.nedt.2019.05.029>.
34. Liaw SY, Wu LT, Lim S, Lau TC. Hybrid models of AI-assisted communication training: Enhancing interprofessional education. *BMC Med Educ*. 2023;23:784. <https://doi.org/10.1186/s12909-023-04245-2>.
35. Wang MH, Jiang X, Zeng P, Li X, Chong KKL. Balancing accuracy and user satisfaction: The role of prompt engineering in AI-driven healthcare solutions. *Front Artif Intell*. 2025. <https://doi.org/10.3389/frai.2025.1517918>.
36. Ghasemi S, Dashti M, Vargas N. Revolutionizing head and neck reconstruction: The integration of artificial intelligence and virtual reality in surgical practice. *Preprints org*. 2025. <https://doi.org/10.1101/2025.03.0087>.
37. Alimour SA, Alrabeei M. A novel model for digital twins in mental health: The Biopsychosocial AI-Driven Digital Twin (BADT) Framework. 2024 11th International Conference on eHealth and Bioengineering. 2024. <https://doi.org/10.1109/eHB.2024.10883917>

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [FinalQuestionnaireEvaluationofAI.pdf](#)